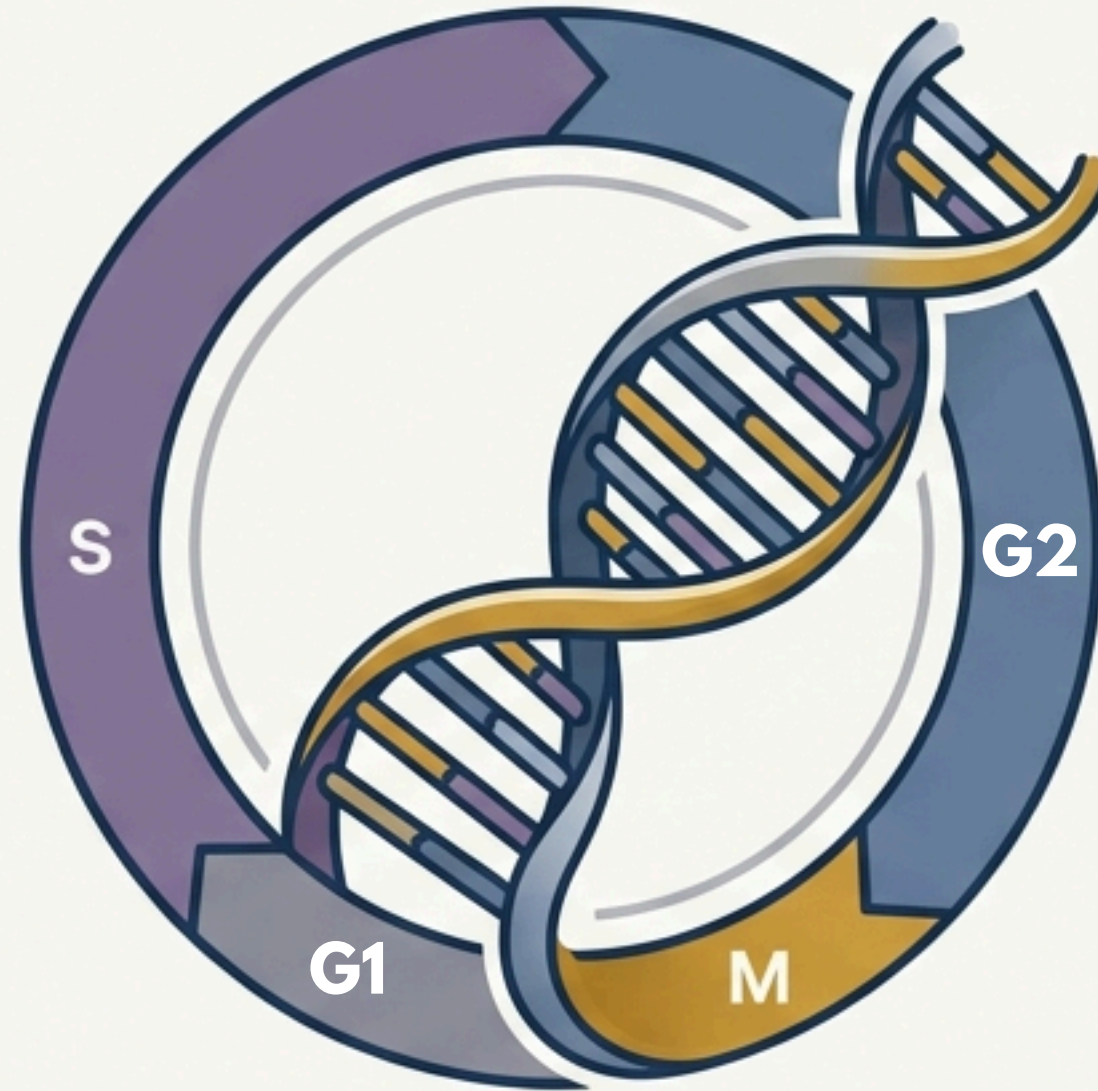


# Cell Cycle and Cell Division

NCERT Class 11 | Chapter 10



An Exam-Oriented Masterclass for NEET-UG

Strictly Aligned with the Latest NCERT & NEET Syllabus

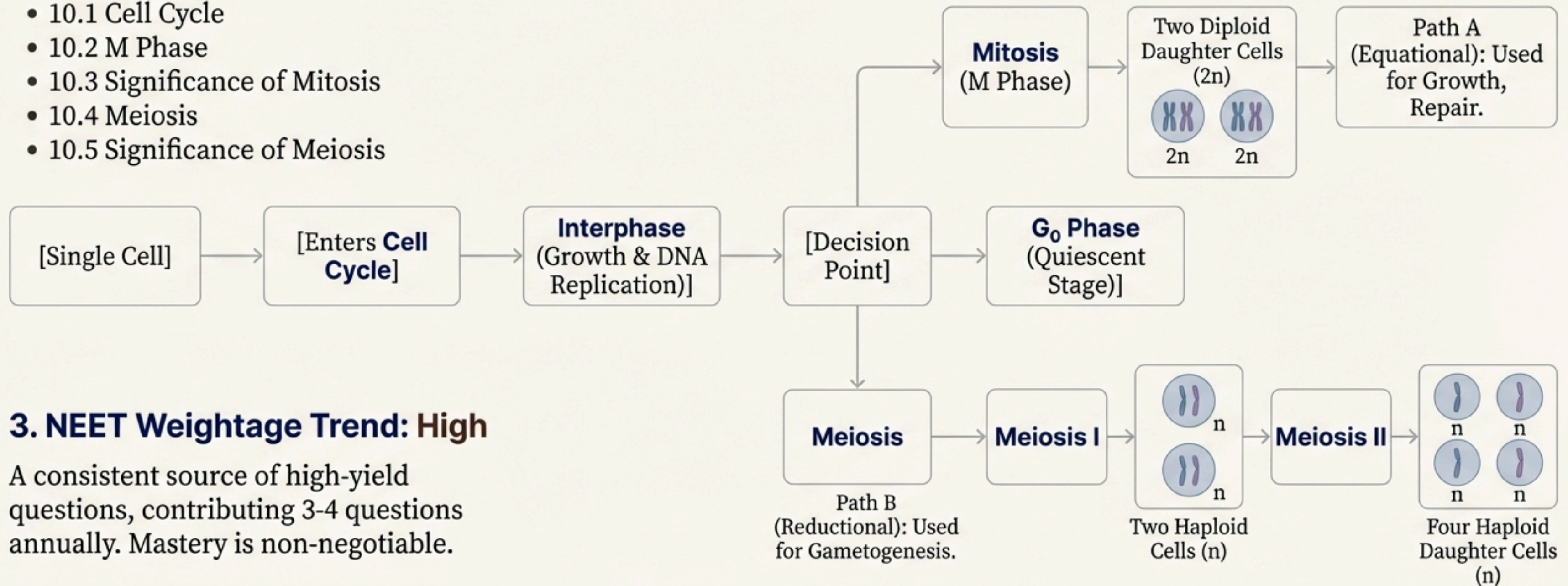


# Chapter Overview: The Blueprint of Cellular Reproduction

## 1. NCERT Syllabus Mapping

- 10.1 Cell Cycle
- 10.2 M Phase
- 10.3 Significance of Mitosis
- 10.4 Meiosis
- 10.5 Significance of Meiosis

## 2. High-Level Flow of Concepts



## 3. NEET Weightage Trend: High

A consistent source of high-yield questions, contributing 3-4 questions annually. Mastery is non-negotiable.



# 10.1 The Cell Cycle: An Overview

## Definition: Cell Cycle

The sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell and eventually divides into two daughter cells. Cell division, DNA replication, and cell growth must occur in a coordinated way.

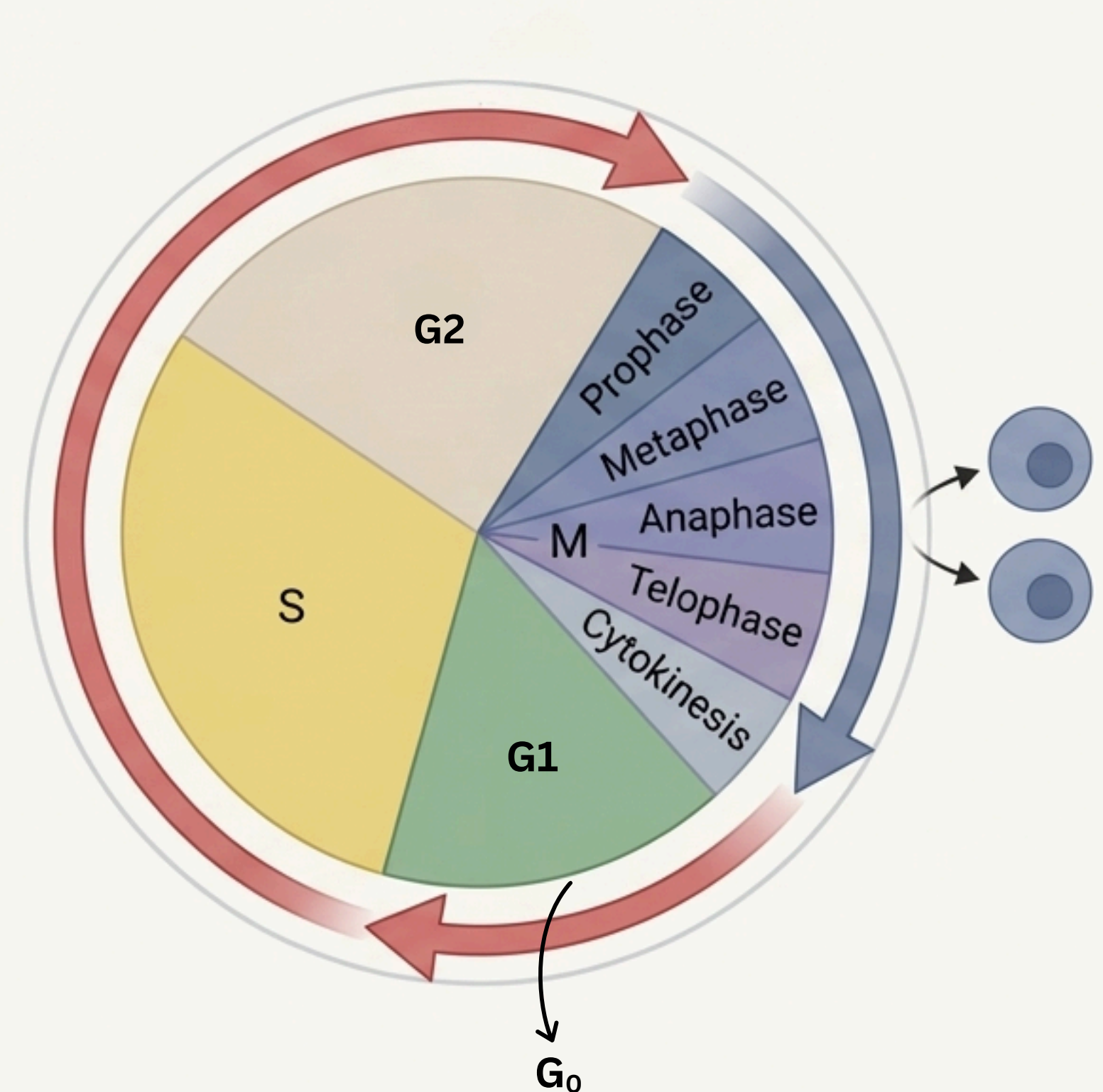
## The Two Basic Phases:

### 1. Interphase:

- The phase between two successive M phases.
- Lasts more than 95% of the cell cycle duration in a typical 24-hour human cell cycle.
- Though called the 'resting phase,' it is a period of intense cell growth and DNA replication.

### 2. M Phase (Mitosis Phase):

- The phase of actual cell division.
- Lasts for only about an hour in a human cell.





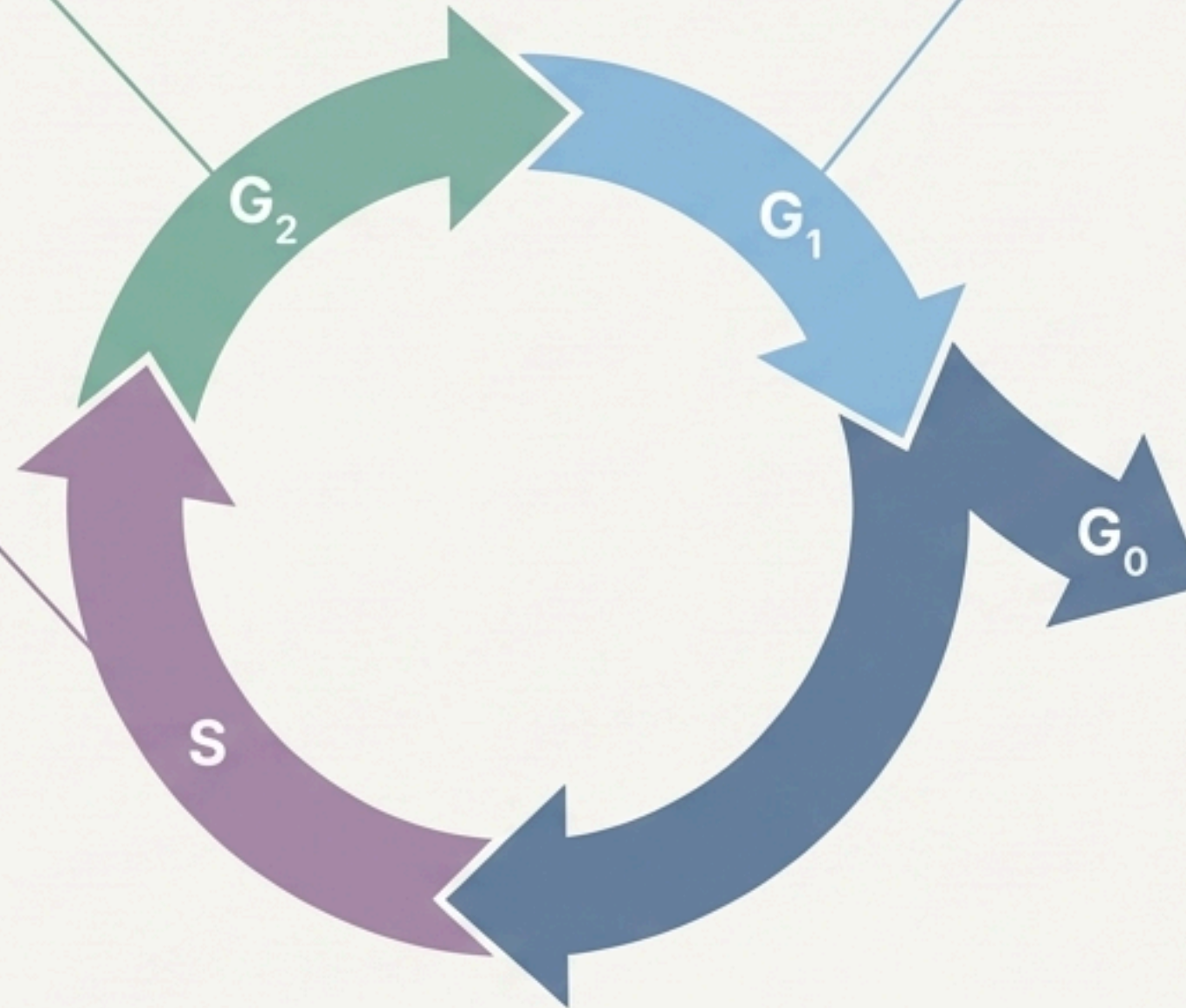
# Interphase: The Preparatory Stages

## G<sub>2</sub> Phase (Gap 2)

- Proteins are synthesised in preparation for mitosis.
- Cell growth continues.

## S Phase (Synthesis)

- **DNA synthesis or replication takes place.**
- The amount of DNA per cell doubles (e.g., from 2C to 4C).
- **Crucially, the chromosome number remains the same** (e.g., 2n remains 2n).
- In animal cells, the centriole duplicates in the cytoplasm.



## G<sub>1</sub> Phase (Gap 1)

- Interval between mitosis and the initiation of DNA replication.
- The cell is metabolically active and grows continuously.

## G<sub>0</sub> Stage (Quiescent Stage)

- Cells that do not divide further exit the G<sub>1</sub> phase to enter this inactive stage.
- They remain metabolically active but no longer proliferate unless required.
- Example from NCERT: Heart cells.



# 10.2 M Phase: Mitosis (Equational Division)

## Definition:

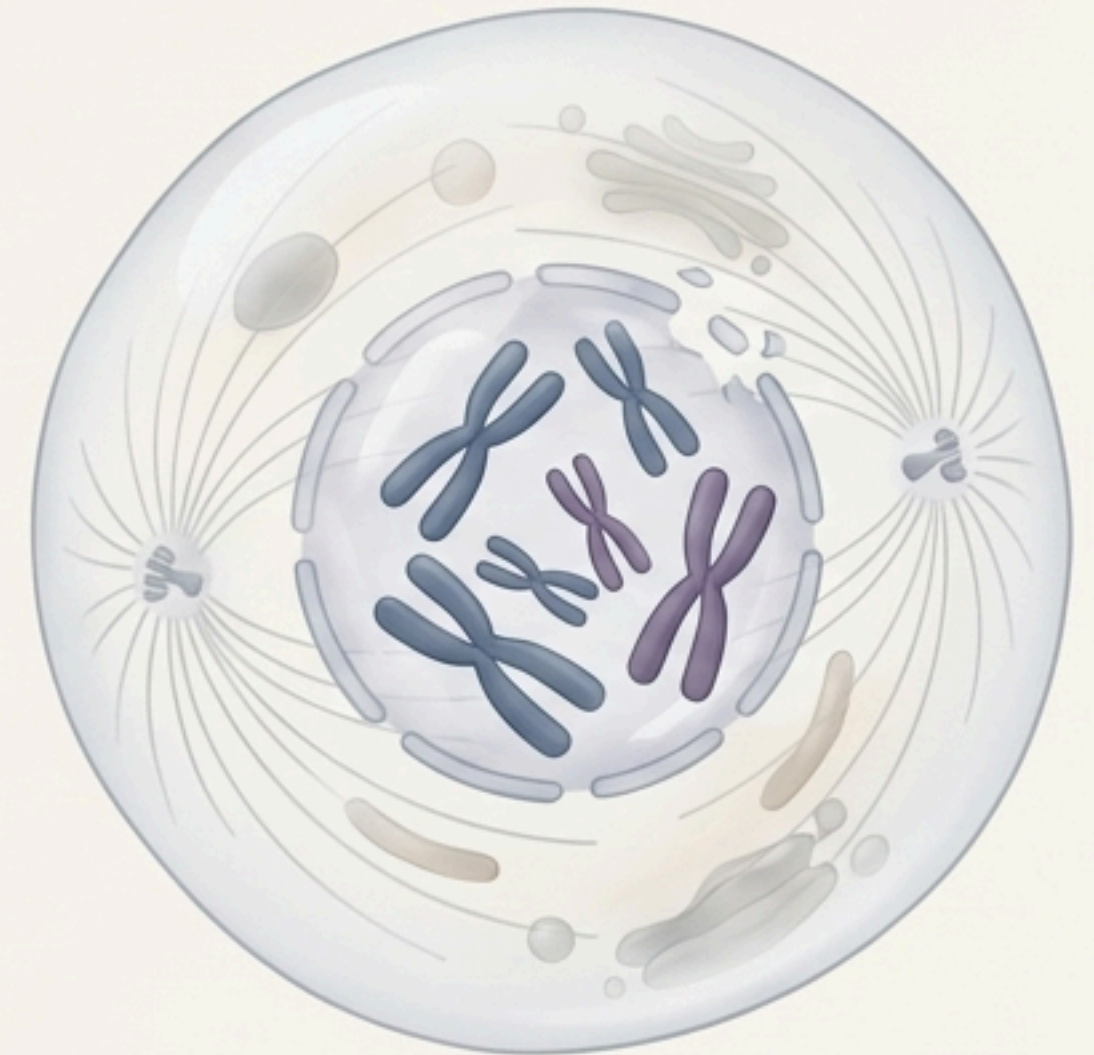
A division where the number of chromosomes in the parent and progeny cells remains the same. Also called **equational division**.

Occurs in diploid somatic cells in animals.

Occurs in both haploid and diploid cells in plants.

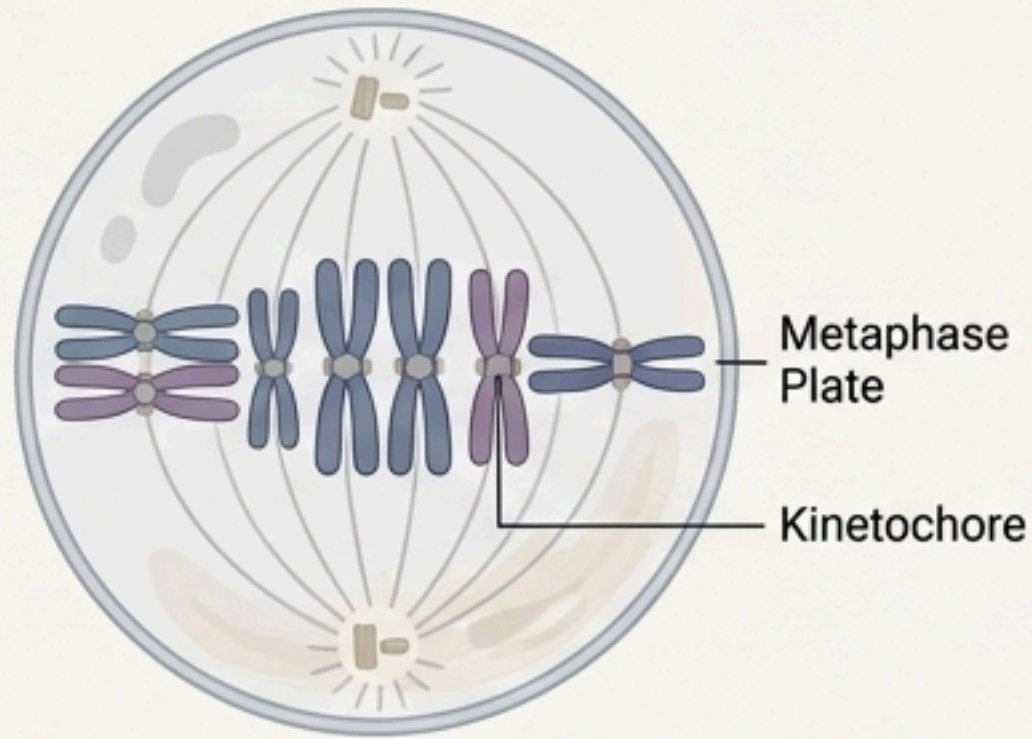
## Karyokinesis Stage 1: Prophase

- Follows the S and G<sub>2</sub> phases.
- **Key Events:**
  - **Initiation of condensation** of chromosomal material, which becomes untangled.
  - Chromosomes become visible, composed of two chromatids attached at the centromere.
  - The centrosome (duplicated during S phase) begins to move towards opposite poles.
  - Each centrosome radiates microtubules called **asters**. The two asters plus spindle fibres form the **mitotic apparatus**.
  - **At the end of Prophase:** The golgi complex, endoplasmic reticulum, nucleolus, and nuclear envelope are no longer visible.



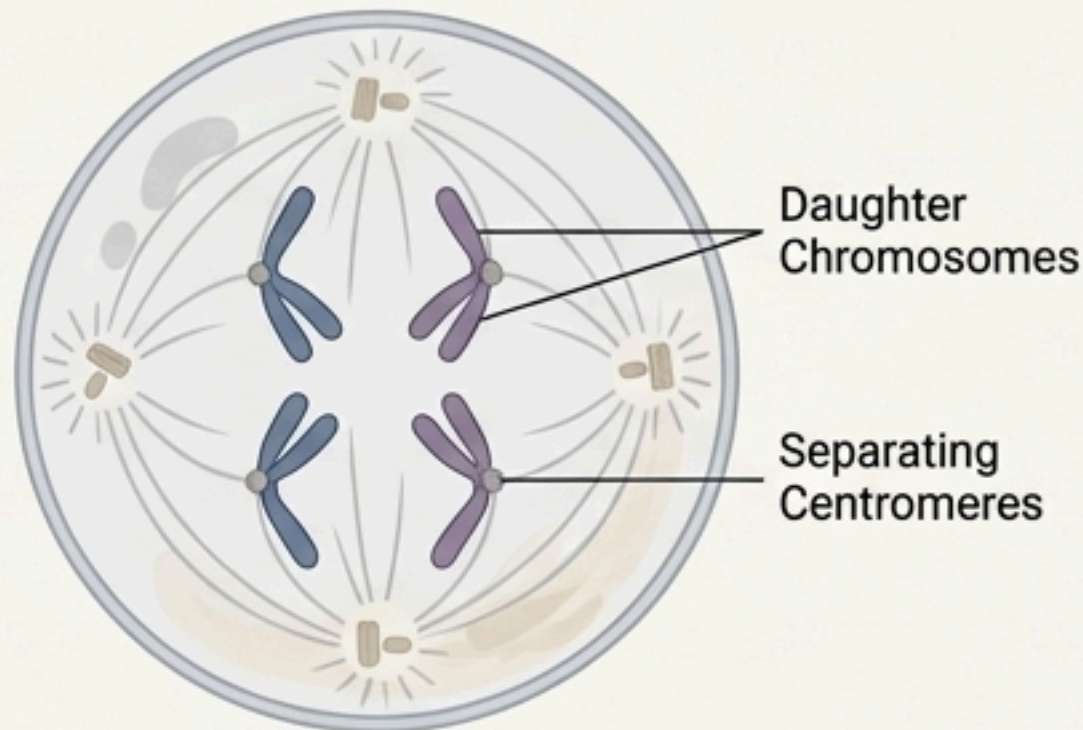


# Mitosis: Alignment (Metaphase) and Separation (Anaphase)



## Karyokinesis Stage 2: Metaphase

- Begins with the complete disintegration of the nuclear envelope.
- Chromosome condensation is complete; morphology is most easily studied at this stage.
- Small disc-shaped structures on the centromeres called kinetochores serve as sites for spindle fibre attachment.
- Key Event: All chromosomes align at the cell's equator on the metaphase plate.



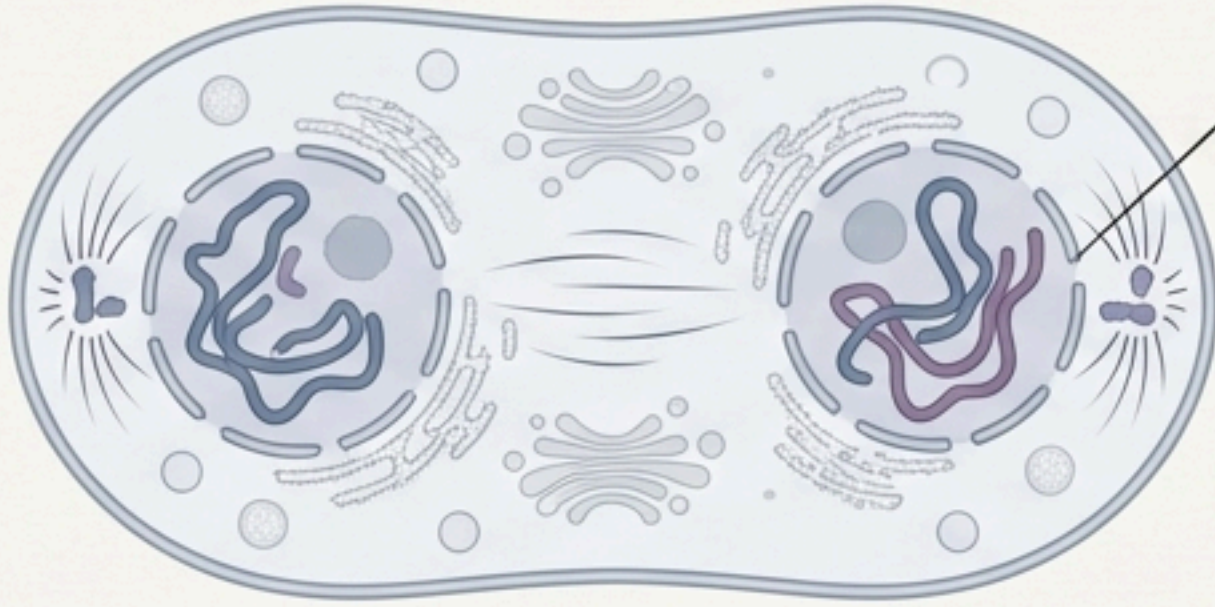
## Karyokinesis Stage 3: Anaphase

- Key Events:
  - Centromeres split simultaneously.
  - The two daughter chromatids separate and are now called daughter chromosomes.
  - These daughter chromosomes migrate towards the two opposite poles, with the centromere at the leading edge.



# Mitosis: Reformation (Telophase) and Division (Cytokinesis)

## Karyokinesis Stage 4: Telophase

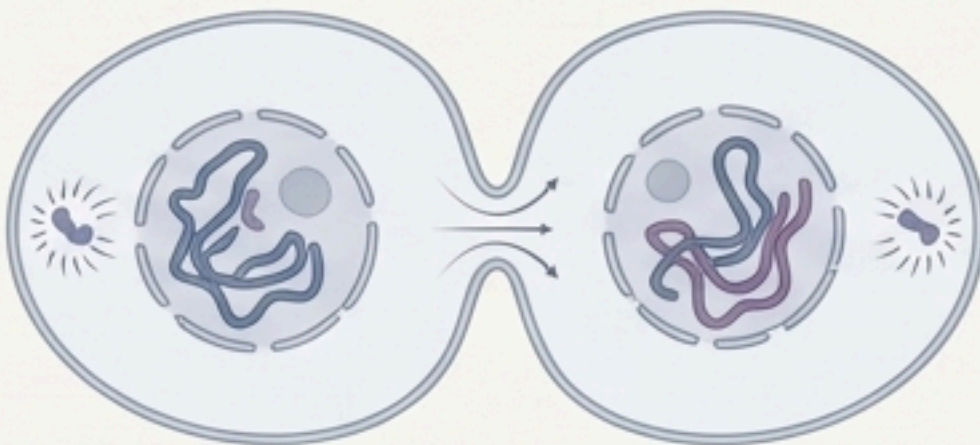


The chromosomes cluster at their respective poles, decondense, and lose their individual identity.

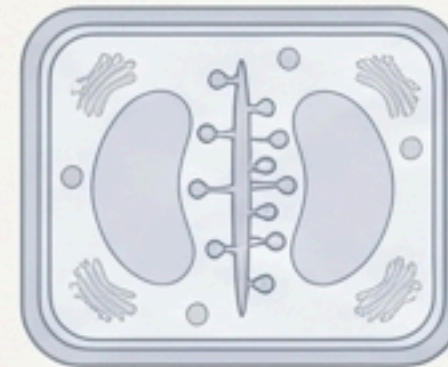
### Key Events:

- The nuclear envelope reassembles around the chromosome clusters at each pole.
- The nucleolus, golgi complex, and ER reform.

## Cytokinesis: Division of the Cytoplasm



**Animal Cells:** A furrow in the plasma membrane deepens to divide the cell.



**Plant Cells:** A cell-plate forms in the center and grows outward. This represents the middle lamella.

**Syncytium:** If karyokinesis is not followed by cytokinesis, a multinucleate condition arises (e.g., liquid endosperm in coconut).



# 10.4 Meiosis I: Reductional Division & The Complexity of Prophase I

## Meiosis Overview

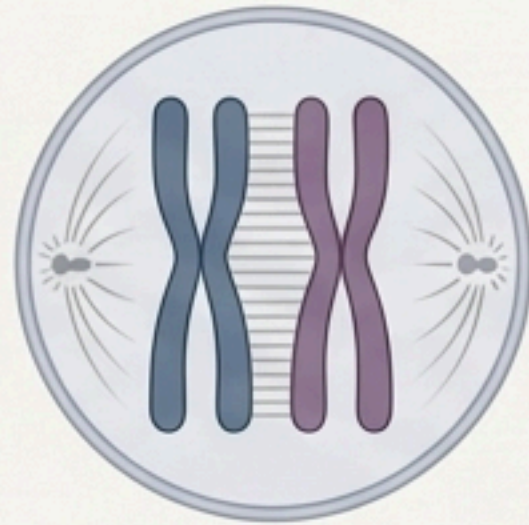
- A specialised division that reduces the chromosome number by half.
- Involves two sequential cycles (Meiosis I & II) but only a single cycle of DNA replication.
- Involves pairing of homologous chromosomes and recombination.

## The Five Sub-stages of Prophase I



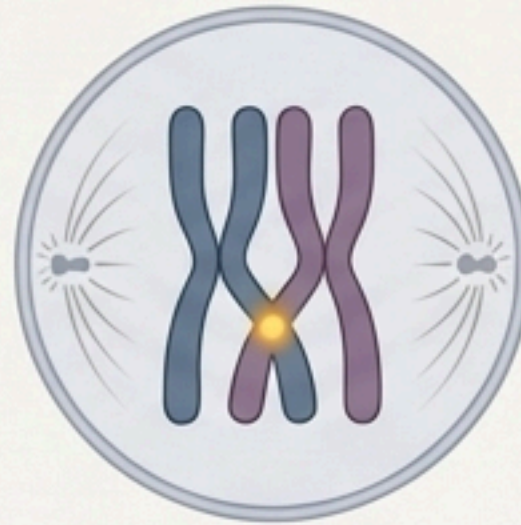
### 1. Leptotene

★ Chromosomes gradually become visible and continue to compact.



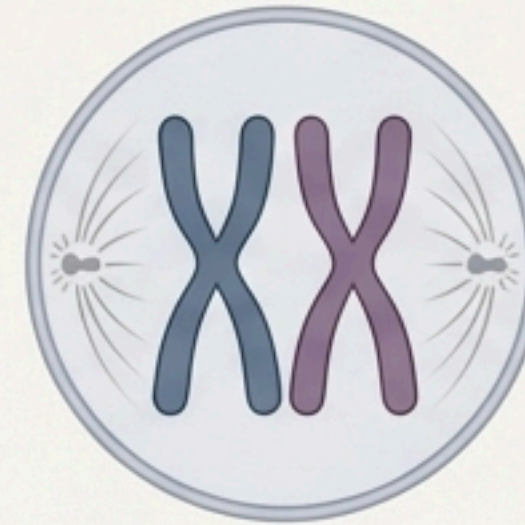
### 2. Zygotene

★ Pairing of homologous chromosomes (**synapsis**) begins, facilitated by the **synaptonemal complex**. Paired chromosomes are called **bivalents** or **tetrads**.



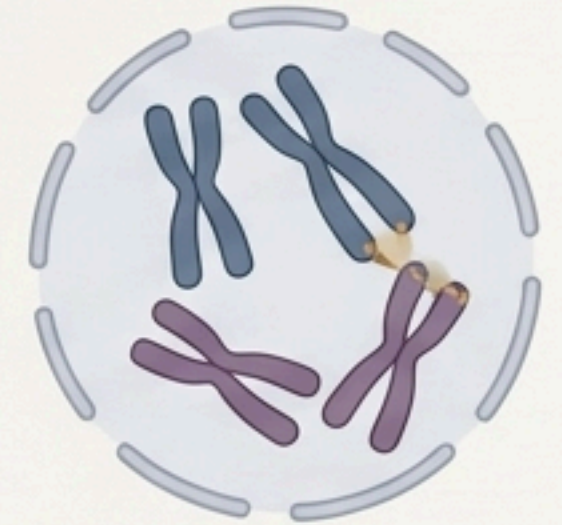
### 3. Pachytene

★ **Crossing over** (exchange of genetic material between non-sister chromatids of homologous chromosomes) occurs at sites called **recombination nodules**. This is an enzyme-mediated process involving **recombinase**.



### 4. Diplotene

★ The synaptonemal complex dissolves. Homologous chromosomes separate, except at the sites of crossover, forming X-shaped structures called **chiasmata**.



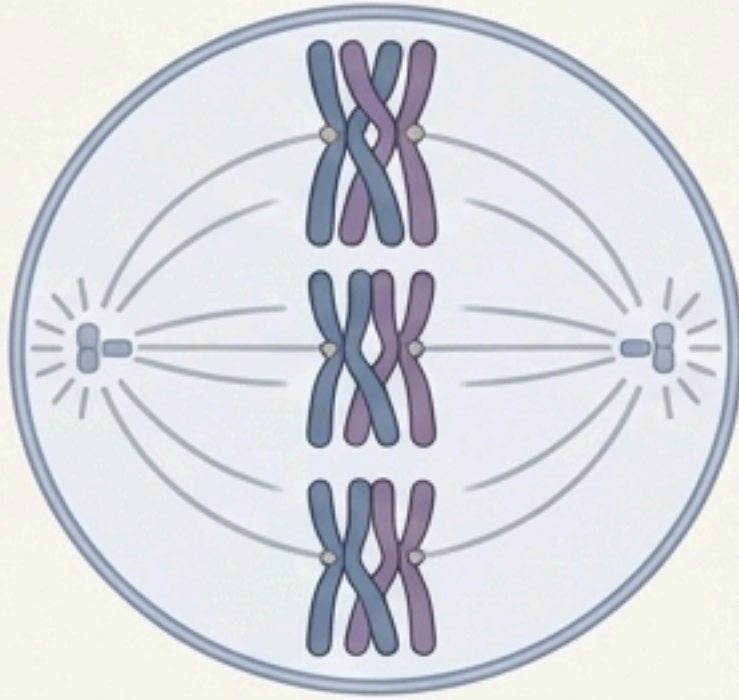
### 5. Diakinesis

★ Marked by the **terminalisation of chiasmata**. Chromosomes are fully condensed. The nucleolus disappears and the nuclear envelope breaks down.



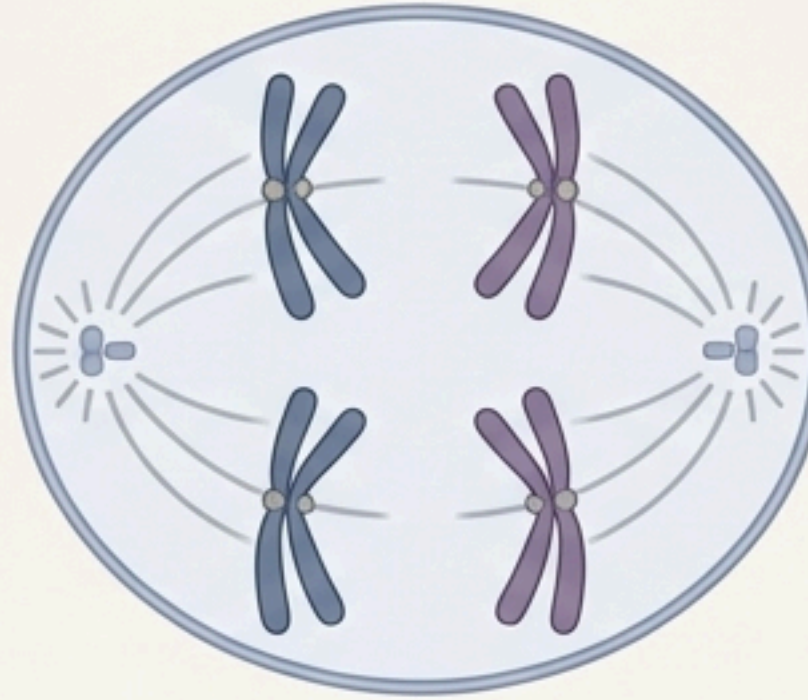
# Completing Meiosis I: Separation of Homologous Chromosomes

## Metaphase I



- **Bivalent chromosomes** align on the equatorial plate.
- Microtubules from opposite poles attach to the kinetochores of **homologous chromosomes**.

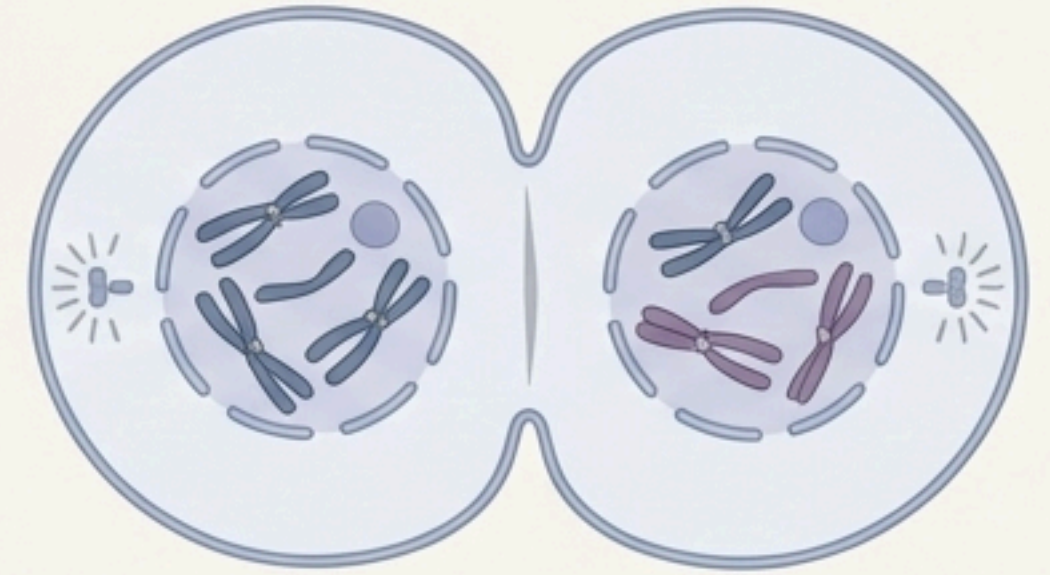
## Anaphase I



### The Key Event of Meiosis I:

- The homologous chromosomes **separate**.
- Sister chromatids remain associated at their centromeres.
- This step reduces the chromosome number at each pole.

## Telophase I & Interkinesis



### Telophase I:

- The nuclear membrane and nucleolus reappear.
- Cytokinesis follows, forming a **dyad of cells** (two haploid cells).

**Interkinesis:** The stage between the two meiotic divisions. It is generally short-lived.  
★ **There is no replication of DNA during interkinesis.**



## 10.4.2 Meiosis II: The Equational Stage

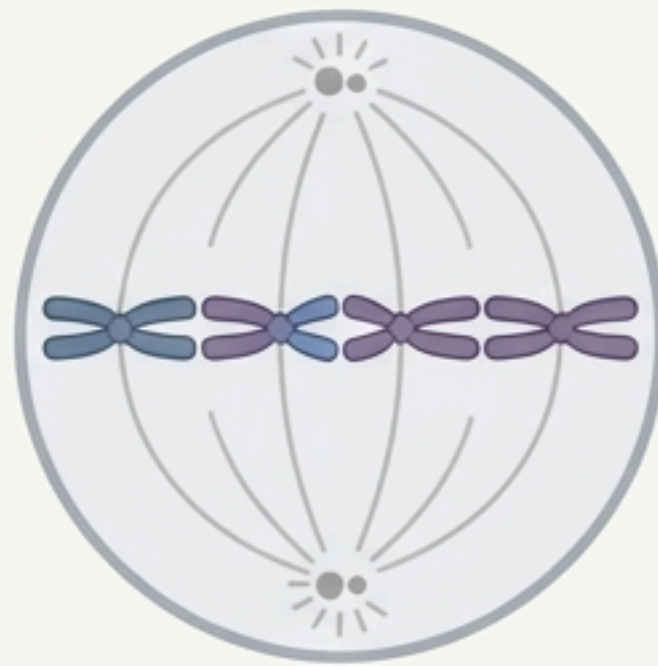
Note: Meiosis II resembles a normal mitosis and is initiated immediately after cytokinesis.

### Prophase II



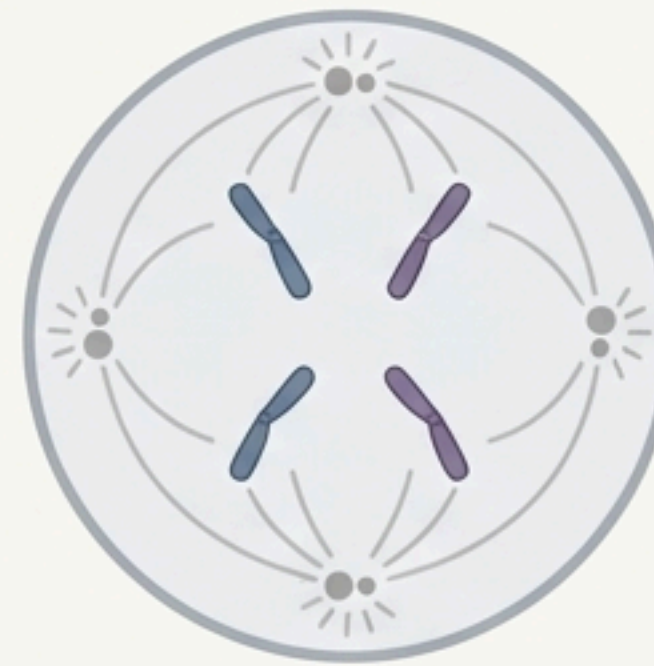
The nuclear membrane disappears. The chromosomes again become compact.

### Metaphase II



The chromosomes align at the equator. Microtubules from opposite poles attach to the kinetochores of **sister chromatids**.

### Anaphase II



Begins with the **simultaneous splitting of the centromere** of each chromosome. This allows the **sister chromatids to separate** and move toward opposite poles.

### Telophase II & Final Product



Meiosis ends. The two groups of chromosomes get enclosed by a nuclear envelope. Cytokinesis follows, resulting in the formation of a **tetrad of cells** (i.e., four haploid daughter cells).



## ★ NCERT Line Emphasis: Direct NEET Pickups

⚠ **NCERT Line:** 'The interphase lasts more than 95% of the duration of cell cycle.'

★ **Direct NEET Pickup:** Often tested as a percentage or duration comparison question.

⚠ **NCERT Line:** 'During this time [S phase] the amount of DNA per cell doubles... However, there is no increase in the chromosome number...'

★ **Direct NEET Pickup:** The basis for all C and N value problems. Critical concept.

⚠ **NCERT Line:** 'This then, is the stage [Metaphase] at which morphology of chromosomes is most easily studied.'

★ **Direct NEET Pickup:** A frequent one-liner factual recall question.

⚠ **NCERT Line:** 'Crossing over is the exchange of genetic material between two **homologous chromosomes...between non-sister chromatids...**'

★ **Direct NEET Pickup:** Precision is key. Examiners will create options with 'sister chromatids' or 'non-homologous chromosomes' to trap students.

⚠ **NCERT Line:** 'The stage between the two meiotic divisions is called interkinesis... There is **no replication of DNA** during interkinesis.'

★ **Direct NEET Pickup:** A critical point of difference from the interphase preceding mitosis or Meiosis I.



# Comparative Analysis: Mitosis vs. Meiosis

| Parameter         | Mitosis                                                               | Meiosis                                                                                              |
|-------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Occurs in         | Somatic cells; both haploid & diploid cells in plants                 | Germline cells (for gametogenesis)                                                                   |
| Division Type     | Equational Division                                                   | Reductional (Meiosis I) & Equational (Meiosis II)                                                    |
| No. of Divisions  | One                                                                   | Two (Meiosis I and II)                                                                               |
| Synapsis          | Absent                                                                | Occurs during Zygotene of Prophase I                                                                 |
| Crossing Over     | Absent                                                                | Occurs during Pachytene of Prophase I                                                                |
| Anaphase Event    | Sister chromatids separate                                            | <b>Anaphase I:</b> Homologous chromosomes separate<br><b>Anaphase II:</b> Sister chromatids separate |
| Chromosome No.    | Daughter cells have the same number as parent ( $2n \rightarrow 2n$ ) | Daughter cells have half the number of parent ( $2n \rightarrow n$ )                                 |
| No. of Daughters  | Two diploid cells                                                     | Four haploid cells                                                                                   |
| Genetic Variation | Does not introduce variation                                          | Introduces significant variation via crossing over                                                   |
| Significance      | Growth, cell repair, restoring nucleo-cytoplasmic ratio               | Conservation of chromosome number across generations, creating genetic variability for evolution     |



# NEET-UG Focus: Common Traps & Confused Terms

## 1. Previous Year Question Themes

- Identifying events of G1, S, and G2 phases.
- Correct sequence of Prophase I stages (Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis).
- Events of Pachytene (Crossing over) & Diplotene (Chiasmata).
- The fundamental difference between Anaphase of Mitosis, Anaphase I, and Anaphase II.
- Calculating Chromosome (N) and DNA content (C) at various stages.

## 2. Common Traps & Misconceptions

- **⚠ Trap:** Confusing 'Chromosome Number' (count of centromeres) with 'DNA Content'. After S phase, chromosome number is  $2n$ , but DNA content is  $4C$ .
- **⚠ Trap:** Assuming DNA replication occurs during interkinesis. **FALSE.** It has no S phase.
- **⚠ Trap:** Thinking crossing over occurs between sister chromatids. **FALSE.** It occurs between non-sister chromatids of homologous chromosomes.

## 3. Frequently Confused Terms

- **Centromere:** Holds sister chromatids together.
- **Kinetochores:** Disc-shaped protein structure on the centromere where spindle fibres attach.
- **Bivalent:** A pair of synapsed homologous chromosomes.
- **Tetrad:** The four chromatids of a bivalent, clearly visible in Pachytene.



# Conceptual Check: Assertion–Reason & Statement Analysis

**Assertion (A):** Mitosis is called equational division.

**Reason (R):** The number of chromosomes in the parent and progeny cells is the same.

**Analysis:** A is true, R is true, and R is the correct explanation of A.

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**Assertion (A):** Meiosis increases the genetic variability in the population of organisms.

**Reason (R):** Meiosis involves crossing over, which leads to recombination of genetic material.

**Analysis:** A is true, R is true, and R is the correct explanation of A.

---

## Statement Analysis (True/False):

**Statement 1:** DNA replication occurs during interkinesis.

**Analysis: False.** The source explicitly states, ‘There is no replication of DNA during interkinesis.’

**Statement 2:** In Anaphase I of meiosis, centromeres split and sister chromatids separate.

**Analysis: False.** Homologous chromosomes separate; sister chromatids remain associated at their centromeres.

**Statement 3:** The formation of a cell plate during cytokinesis is characteristic of plant cells.

**Analysis: True.** This is due to the presence of a relatively inextensible cell wall.

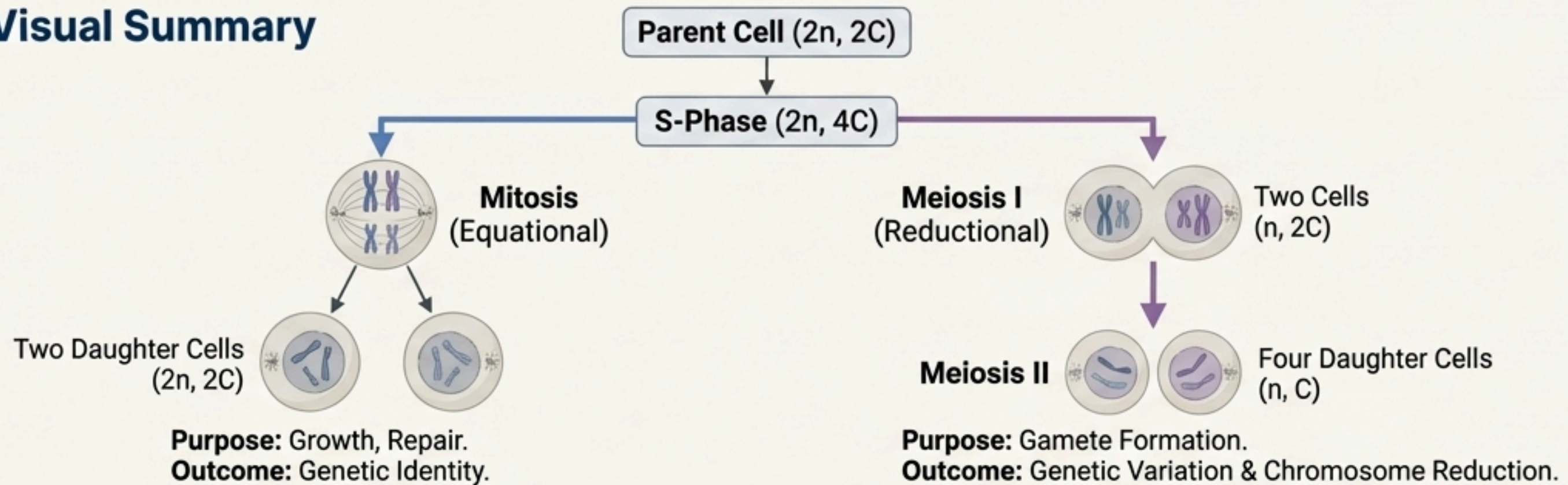


# Final Chapter Snapshot: The Flow of Genetic Material

## Rapid Revision Keywords & Sequences

- **Cell Cycle:** Interphase (G1, S, G2) → M Phase (Prophase, Metaphase, Anaphase, Telophase) → Cytokinesis.
- **Prophase I Sequence:** Leptotene → Zygotene (Synapsis) → Pachytene (Crossing Over) → Diplotene (Chiasmata) → Diakinesis (Terminalisation).
- **Key Distinctions:**
  - **Mitotic Anaphase:** Centromeres split, Sister Chromatids separate.
  - **Meiotic Anaphase I:** Homologous Chromosomes separate.
  - **Meiotic Anaphase II:** Centromeres split, Sister Chromatids separate.

## One-Slide Visual Summary



**'If you remember only this much, you can solve most questions:'**

Mitosis separates identical sisters. Meiosis I separates homologous partners.  
Meiosis II separates the sisters (now non-identical due to crossing over).